DELHI PUBLIC SCHOOL, BOKARO STEEL CITY

Biology — Theory Paper

Time: 1 Hour 15 Minutes | Maximum Marks: 26

1 Fig. 1.1 shows the human heart and the main blood vessels. The functions of the parts of the heart and some of the blood vessels are given in Table 1.1. Complete Table 1.1 by naming the part that separates the two sides of the heart, the valve between the atrium and the ventricle, the vessel carrying oxygenated blood to the body, and the vessel returning deoxygenated blood to the heart. [4]

2 The nervous system coordinates the responses of animals to changes in their environment. Fig. 2.1 shows the arrangement of the nervous system in a mammal. Complete Fig. 2.1 by writing the names of the two divisions of the nervous system and the structure that runs inside the vertebral column. [3]

3 Catalase is an enzyme that breaks down hydrogen peroxide inside cells. Some dogs have an inherited condition in which catalase is not produced. This condition is known as acatalasia and is caused by a mutation in the gene for catalase.

(a) Define the terms *gene* and *gene mutation*. [2]

(b) Fig. 3.1 shows the inheritance of acatalasia in a family of dogs, where the normal allele is represented by B and the mutant allele by b. State the genotypes of the dogs identified as 1, 2 and 3 in Fig. 3.1. [3]

4 *Rhabdostyla* is a single-celled organism that has no cell wall and no chlorophyll. It lives in freshwater habitats such as ponds, lakes and rivers.

(a) Gases are exchanged across the cell membrane of *Rhabdostyla*. Name the gas produced by *Rhabdostyla*, the process that produces the gas, and the method of removal of the gas. [3]

(b) Explain, using the term water potential, why *Rhabdostyla* needs to remove excess water. [3]

5 Fig. 5.1 shows the mass of wild fish caught worldwide between 1950 and 2012. Describe the changes in the mass of wild fish caught between 1950 and 2012. You will gain credit if you use data from Fig. 5.1. [4]

6 Define transpiration and explain how wind speed and humidity each affect the rate at which it occurs in a leafy plant. [4]